EXP 3

AIM: To compare a simplified free-space path-loss estimate across representative sub-1 GHz, midband and mmWave frequencies, and examine how distance and frequency reuse affect received signal and a simple interference measure

Procedure:

1. Run the script and inspect the path-loss and received-power plots.

2. Record the estimated received power at 0.5 km for each frequency.

3. Inspect the reuse-SIR plot and compare the baseline N values.

4. Change transmit power, distance range, path-loss exponent or reuse sizes; rerun and record the effect.

5. Explain why frequency reuse changes the simplified co-channel SIR and why the freespace link budget is not a complete coverage prediction.

CODE:

```
clear;

clc;

close all;

%% 1. Frequency Bands and Link-Budget Parameters

freqGHz = [0.8 2.6 26];

freqMHz = freqGHz * 1000;

bandName = ["Low-band"; "Mid-band"; "mmWave"];

txPowerdBm = 33;

gtDbi = 2;

grDbi = 1;
```

```
distanceKm = logspace(log10(0.1), log10(3), 200);
distanceAtPointKm = 0.8;
%% 2. Free-Space Path Loss Function
% Distance in km and frequency in MHz
fspl = @(fMHz, dKm) 32.44 + ...
   20*log10(fMHz) + 20*log10(dKm);
%% 3. Calculate Path Loss and Received Power
lossAtPoint = fspl(freqMHz, distanceAtPointKm);
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
linkTable = table(bandName, freqGHz(:), lossAtPoint(:), ...
   rxAtPoint(:), ...
   'VariableNames', {'Band', 'Frequency_GHz', ...
   'FSPL_dB', 'EstimatedRx_dBm'});
disp('LINK BUDGET RESULTS:');
disp(linkTable);
%% 4. Plot Free-Space Path Loss and Received Power
figure('Color', 'w');
tiledlayout(1, 2);
% Free-space path loss
nexttile;
hold on;
grid on;
for k = 1:numel(freqMHz)
   plot(distanceKm, fspl(freqMHz(k), distanceKm), ...
      'LineWidth', 1.8);
end
set(gca, 'XScale', 'log');
```

```
xlabel('Distance (km)');
ylabel('Path Loss (dB)');
legend(bandName, 'Location', 'northwest');
title('Free-Space Path Loss');
% Estimated received power
nexttile;
hold on;
grid on;
for k = 1:numel(freqMHz)
    rx = txPowerdBm + gtDbi + grDbi ...
        - fspl(freqMHz(k), distanceKm);
    plot(distanceKm, rx, 'LineWidth', 1.8);
end
set(gca, 'XScale', 'log');
xlabel('Distance (km)');
ylabel('Received Power (dBm)');
legend(bandName, 'Location', 'southwest');
title('Estimated Received Power');
%% 5. Co-Channel Reuse and SIR Estimation
cellRadiusKm = 0.6;
reuseN = [1 3 4 7 12];
pathLossExponent = 3.2;
reuseDistance = sqrt(3 .* reuseN) .* cellRadiusKm;
% Simplified first-tier, six-interferer model
sirLinear = (reuseDistance ./ cellRadiusKm) ...
    .^ pathLossExponent ./ 6;
```

```
sirDb = 10 .* log10(sirLinear);
reuseTable = table(reuseN(:), reuseDistance(:), sirDb(:), ...
    'VariableNames', {'ReuseClusterN', ...
    'ApproxInterfererDistance_km', 'ApproxSIR_dB'});
disp('CO-CHANNEL REUSE RESULTS:');
disp(reuseTable);
%% 6. Plot SIR versus Reuse Cluster Size
figure('Color', 'w');
plot(reuseN, sirDb, 'o-', ...
    'LineWidth', 1.8, 'MarkerSize', 7);
grid on;
xlabel('Reuse Cluster Size (N)');
ylabel('Approximate SIR (dB)');
title('Co-Channel SIR versus Reuse Cluster Size');
%% 7. Display Received Power at the Selected Distance
fprintf('\nAt %.2f km, estimated received powers are:\n', ...
    distanceAtPointKm);
powerTable = table(bandName, rxAtPoint(:), ...
    'VariableNames', {'Band', 'ReceivedPower_dBm'});
disp(powerTable);
```

OUTPUT:

MATLAB Drive

Editor — untitled6.m — /MATLAB Drive/untitled6.m

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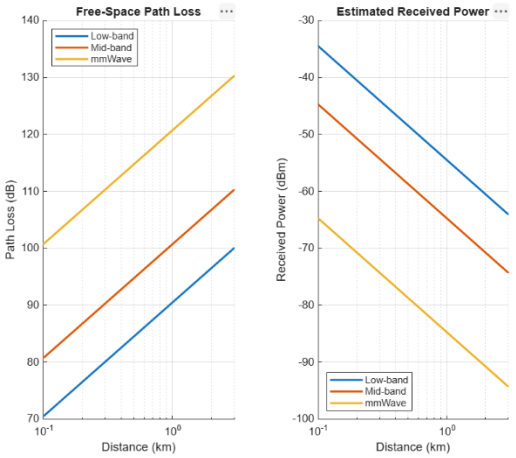
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Apps — Figure 1 — Figure 2



MATLAB Drive

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Apps — Figure 1 — Figure 2

